



TRACK 5 · CYCLONE IMPACT & INFRASTRUCTURE VULNERABILITY FORECASTER

CycloneSentinel

Know which hospitals, substations, telecom towers and homes will fail —
before the cyclone makes landfall.

THE PROBLEM

**Forecasts say *where* the storm goes.
Responders need to know *what breaks*.**

Fani 2019

Puri's power-distribution network largely destroyed; prolonged outages in Puri, Bhubaneswar, Cuttack.

Hudhud 2014

Visakhapatnam's power and telecom networks collapsed for days; ~₹21,900 crore estimated losses.

Amphan 2020

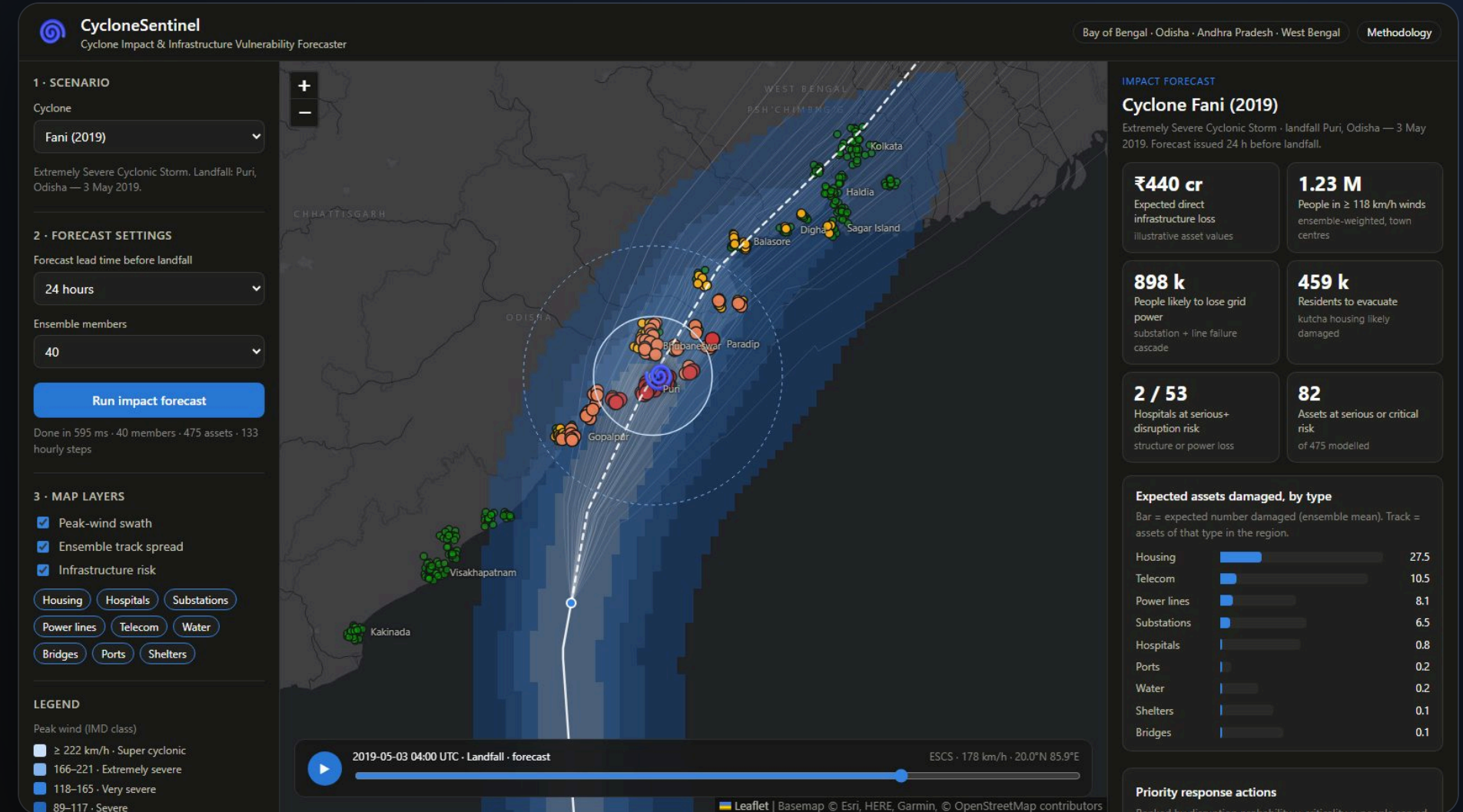
Sundarbans and Kolkata hit by surge, wind and tree-fall — widespread power and telecom failure.

- Warnings are issued per **district**; decisions are made per **asset** — this hospital, that substation.
- Failures **cascade**: an intact hospital goes dark when its feeder substation trips.
- Pre-positioning generators, crews and shelters needs a ranked list **24–72 h ahead**, with honest uncertainty.

THE SOLUTION

Track forecast → asset-level risk → action

- **Hazard** at every asset: wind, storm surge, rainfall flood
- **40-member ensemble** — probabilities, not guesses
- **Fragility curves** for 9 infrastructure classes
- **Cascade** through the power-dependency network
- **Ranked actions**: evacuate, protect, restore, connect
- Runs in the browser in **< 1 s** · zero install



A physics-based impact chain

1 · Track & ensemble

IMD track, hourly. Monte-Carlo members with error $\sigma \approx 45$ km at 24 h, $\pm 12\%$ intensity. Kaplan–DeMaria decay over land.



2 · Hazards

Holland (1980) wind + forward-motion asymmetry + terrain. Shelf-amplified surge. R-CLIPER-style rain $\rightarrow$ flood depth.



3 · Fragility

Lognormal wind & flood curves per asset class (HAZUS-MH derived) $\rightarrow$ P(damage).



4 · Cascade

Substation $\leftarrow$ transmission line; hospitals, water, telecom $\leftarrow$ substation, net of backup $\rightarrow$ P(disruption).



5 · Decisions

KPIs + actions ranked by risk $\times$ criticality $\times$ people served.

475

assets across 34 coastal towns in Odisha, AP, West Bengal

~0.5 s

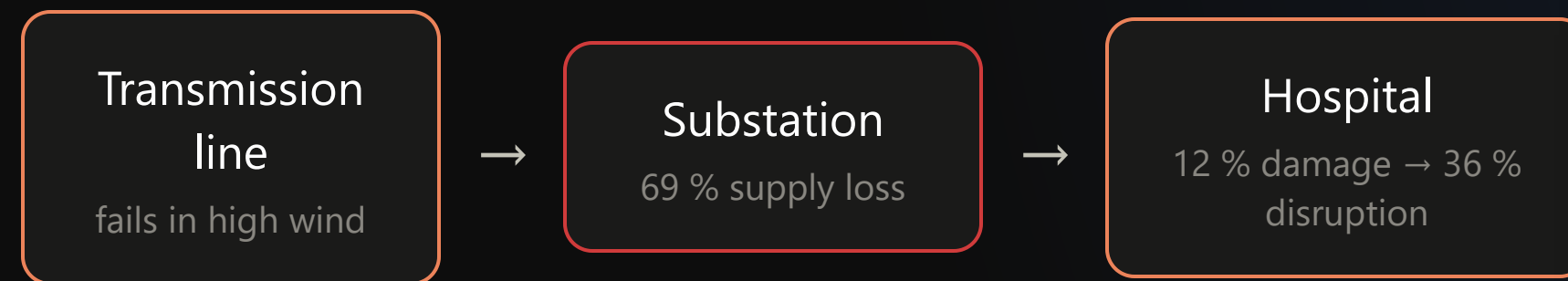
for 40 members $\times$ 475 assets $\times$ 130 hourly steps, in-browser

12–72 h

forecast lead time — uncertainty widens honestly

KEY INSIGHT

Infrastructure fails as a network



- The building survives — the **service** doesn't.
- Backup power is modelled explicitly: hospital DG sets 60 %, telecom batteries 50 %, water 30 %.
- Output is an action, not a map: **"Pre-position DG sets + 72 h fuel at 2 hospitals in Puri — grid feeder likely to trip."**



Hindcast: 4 historical cyclones, forecast issued T–24 h

Cyclone	Actual landfall	Model's top-3 worst-hit towns	People likely without power	
Fani (2019)	Puri, Odisha	Puri, Bhubaneswar, Cuttack	9.0 lakh	✓
Amphan (2020)	Sundarbans, West Bengal	Sagar Island, Haldia, Kolkata	7.5 lakh	✓
Phailin (2013)	Gopalpur, Odisha	Berhampur, Gopalpur, Rambha	3.6 lakh	✓
Hudhud (2014)	Visakhapatnam, AP	Visakhapatnam, Vizianagaram, Bheemunipatnam	15.3 lakh	✓

Automated tests also check the physics: peak wind at the radius of maximum wind, right-of-track bias, monotonic fragility, cascade $\geq$ damage, and spread growing with lead time.

Built for the 72 hours before landfall

SDMAs & district collectors

Where to evacuate first, which shelters are themselves exposed, how many people to move.

Power utilities (discoms)

Which substations and line spans will fail → stage crews and mobile transformers in advance.

Hospitals & health dept.

Which hospitals lose grid power → fuel, generators, patient transfers.

Telecom operators

Towers likely to fail → cell-on-wheels and satellite phones for first responders.

Ports & industry

Surge and wind exposure of terminals → suspend operations, secure vessels.

Planners & insurers

What-if storms to stress-test the coast and prioritise retrofits.

From prototype to production

- Live **IMD / JTWC advisories** and ECMWF / NCMRWF ensembles
- Real assets from **OpenStreetMap**, Bhuvan and discom GIS (prototype inventory is synthetic)
- **DEM + coastline** for surge and inundation
- Fragility **calibration** on Fani, Amphan, Hudhud, Biparjoy, Michaung losses
- **SMS / WhatsApp** alerts to control rooms

Stack

- Vanilla JavaScript model — no server, works offline
- Leaflet map · GitHub Pages hosting
- Node test runner — physics + hindcast tests
- Playwright — automated demo recording

Try it

blop77.github.io/cyclone-sentinel
github.com/Blop77/cyclone-sentinel